

# Page 1 — TNG Shopper

## TNG Shopper — production multi-agent system | Co-founder CEO/CPO + Lead Product Engineer | Jan 2024 – present

### Identity

TNG Shopper is an AI content engine for multi-location brands.

### Architecture (the 7-node DAG)

7-node multi-agent DAG at manzela.github.io/pipeline-observatory/#dag. Each node: deterministic gate paired with probabilistic agent. Six nodes in production (brief intake, retrieval, draft generation, O-R-A-V deterministic gating, schema injection, distribution). Seventh node (reinforcement learning on O-R-A-V signals) in Langfuse R&D. Stack: Google ADK, Vertex AI (Gemini 3.1 Flash-Lite + Gemma 4 26B-A4B-it MoE), Modal serverless + vLLM fallback, Langfuse, Pydantic, OPA, asyncio. Team of 2 over 18 months; Daniel = ~74% of commits.

### Operational numbers

| Metric                     | Value                                    | Verification                           |
|----------------------------|------------------------------------------|----------------------------------------|
| Enterprise customers       | 11                                       | pipeline-observatory/case-studies.html |
| Countries                  | 5 (live: ES, PT, IL; historical: US, MX) | pipeline-observatory/case-studies.html |
| Pages per month            | ~10.5M                                   | pipeline-observatory/index.html        |
| Agent operations per month | ~73.5M                                   | pipeline-observatory/index.html        |
| Cost per page              | \$0.0006                                 | pipeline-observatory/index.html        |
| O-R-A-V pass rate          | 68.9% (by design)                        | pipeline-observatory/index.html        |
| Managed sites              | 234                                      | pipeline-observatory/case-studies.html |

### Role spectrum

Full founder spectrum: ideation, team build, product, engineering, QA, design, sales, ops, research, optimization. Sole engineer on front-end platform (shipped in under a month; demo:youtu.be/czZsPbylC1M). Lead Product Engineer on production pipeline (~74% of commits). Lead on WordPress multisite (team of 3). Closed each enterprise contract personally.

### Elysium (within TNG, 2024–2025)

Research line for physical-context AI in retail demand prediction. Fused search-intent, geo-coordinates, and foot-traffic data. 11 specialized agents. 15 patent claims. Validated across 15,600+ store locations. Paused — folded into main pipeline. Surface: [manzela.github.io/Manzela/elysium/](https://manzela.github.io/Manzela/elysium/).

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## Page 2 — Atelier

**Status — as of 2026-05-19: Week 1 of 3 in sprint to Google for Startups AI Agents Challenge 2026. Submission target 2026-06-03 noon (2 days before 2026-06-05 deadline). Live infrastructure ([atelier.dev](https://atelier.dev), [docs.atelier.dev](https://docs.atelier.dev), [bench.atelier.dev](https://bench.atelier.dev), [calibration.atelier.dev](https://calibration.atelier.dev), [status.atelier.dev](https://status.atelier.dev)) (targeted by 2026-06-03).**

**Atelier — autonomous design agent | Solo architect + lead engineer | May 2026 – present**

### Identity

Atelier converges autonomously to multi-axis judged criteria, terminating on provable quality gates — unlike current tools which generate and stop. Google for Startups AI Agents Challenge 2026 submission, targeting 2026-06-03 noon (\$90K prize pool + DeepMind/Google Labs mentorship).

### Architecture

Three layers. **Layer 3 — Pre-Generation Intake Protocol (PIP)** runs adaptive Q&A mapped 1:1 to documented coding-agent failure patterns; output is an immutable BriefSpec the agent commits to and a seeded DECISIONS.md. **Layer 2 — Campaign Orchestrator** handles cross-surface coherence across multi-session campaigns via Cloud Scheduler + Cloud Tasks. **Layer 1 — Atomic 8-node DAG** runs per-surface: Brief Parser → Source Resolver → EvoDesign LoopAgent → Final Validator + A2UI Renderer. Inside the LoopAgent: Generator (ParallelAgent K=6 candidates via Stitch MCP) → Constitutional Self-Critique → Deterministic Gate (parallel × 6 axes — Lighthouse, axe, visual-diff, token-fidelity, responsive, semantics) → ConsensusAgent (5 specialized rubric judges with DEMAS-D Per-axis Provenance Matrix) → Fixer (Hebbian mutator via `adk optimize / GEPA`). Built on Google ADK 2.0 Beta + Vertex AI + Stitch MCP + A2UI v0.9. Architecture inspired by the production-validated 7-node Gate-Agent DAG in `agent-dag-pipeline` and the long-running-agent harness pattern from Anthropic's published guidance (Nov 2025).

### 13 novel contributions

- **N1 — DGF-D2C** — Deterministic-Gate-First Design-to-Convergence; improves WebGen-Agent SOTA (51.9%) with preconditions before judge calls.
- **N2 — DEMAS-D** — Per-axis Provenance Matrix judges; each axis sees only relevant ground-truth variables.
- **N3 — PerJudge** — Per-Project DPO Judge with Hebbian Mutator; personalized judges win with 20× fewer examples.
- **N4 — PADI** — Project-Agnostic Descriptor Inference; adapts to any tech stack via `.atelier.yaml`.
- **N5 — EvoDesign** — AlphaEvolve-Inspired Evolutionary K-Candidate Search for UI generation.
- **N6 — CSC-D** — Constitutional Self-Critique for Design against Apple-Grade principles.
- **N7 — A2UI-Native Output** — Renders to React + Flutter + Lit + Angular simultaneously.
- **N8 — Public Judge Calibration Dashboard** — Transparency commitment against calibration drift.
- **N9 — Open Eval Adapters Library** — Apache-2.0 PRs to google/adk-python for WebGen-Bench, Design2Code, Web2Code, ScreenSpot, FrontendBench.
- **N10 — Convergence Spec RFC** — Open standard for convergence criteria, trajectories, calibration drift reporting.
- **N11 — Public Eval Harness** — `bench.atelier.dev` accepts submissions from any vendor.

- **N12 — RLRD** — Recursive Long-Running Discipline; Atelier as reference implementation for multi-day, multi-surface campaigns.
- **N13 — PIP** — Pre-Generation Intake Protocol; adaptive-depth intake mapped to documented agent failure patterns.

5 quantified axes

| Axis                        | MVP target | Baseline       |
|-----------------------------|------------|----------------|
| Convergence quality         | ≥ 95%      | ~0% (no gates) |
| Iterations to convergence   | ≤ 3 loops  | 5–10+          |
| Human-in-loop time          | ≤ 60 sec   | ~5–15 min      |
| Pattern reuse               | ≥ 60%      | 0%             |
| WebGen-Bench (NeurIPS 2025) | ≥ 51 pts   | SOTA: 51.9%    |

Build discipline

\$5K Claude Opus 4.7 MAX via Vertex AI. 3-week sprint (2026-05-15 → 06-04). Lockfile-only installs (slopsquatting defense). Conventional Commits 1.0.0 + commitlint. ADR-driven; DECISIONS.md auto-injected into subagent dispatch; REJECTED.md records failed approaches. Subagent tiers: Opus 4.7 MAX (Orchestrator/Reviewer), Sonnet 4.6 (routine), Haiku 4.5 (lint). Six eval surfaces: smoke, integration, WebGen-Bench 484, calibration golden 100, adversarial 50, designer-in-residence (weekly).

Evidence

- [github.com/Manzela/Atelier](https://github.com/Manzela/Atelier) — repo + README + CHANGELOG + DECISIONS.md
- [atelier.dev](https://atelier.dev) — live agent (targeted by 2026-06-03)

Page 3 — TNG WP Multisite + Antigravity-OS

TNG WP Multisite — production AI-search-aware infrastructure | Lead Product Engineer, team of 3 | 2024 – present

234 managed websites across ES / EN / PT / HE multilingual locales. Sage 10 (Roots/Acorn) on PHP 8.1+ with **7,023 lines of custom theme code** and 10 service providers. Custom sunrise.php for dynamic global blog resolution. **1,056-line JSON-LD schema generator** with Schema.org SpeakableSpecification for voice-readout surfaces. AI-bot-aware robots.txt with explicit ALLOW (OAI-SearchBot, ChatGPT-User, PerplexityBot, ClaudeBot, Applebot, Amazonbot, DuckAssistBot) / BLOCK (GPTBot, anthropic-ai, CCBot, Bytespider, Google-Extended) contracts. 3-mode routing, programmatic site spin-up, multilingual mega-slug single-source-of-truth, Cloudflare CDN, Bing IndexNow integration.

Why this matters for any AI-search-aware lab: few candidates have shipped infrastructure that is literally aware of OAI-SearchBot, ClaudeBot, and PerplexityBot at the robots.txt level. The work is the inverse policy of the personal-site infra (open to AI corpus inclusion) — a deliberate commercial-protection BLOCK list on the TNG side that ALLOWS the corpus inclusion bots and BLOCKS the training bots.

Evidence: [github.com/Manzela/WP-Multisite](https://github.com/Manzela/WP-Multisite) (sanitized public blueprint) + any TNG customer site's /robots.txt + /llms.txt. Canonical private source is tng-shopper/SubDirectory-WP-Master-Template.

## Antigravity-OS — open-source governance kernel for AI agents | Solo author | 2025 – present

An open-source fail-closed safety layer for agent fleets: 9-rule Constitution (Rules 00–08), Cost Guard, Flight Recorder, Self-Healing CI, Dreaming Module for offline agent self-improvement, and drop-in MCP server integration. PyPI: `ag-os`. MIT-licensed. v1.4.0 ('DreamEngine Convergence') shipped 2026-05-08.

Evidence: [pypi.org/project/ag-os/](https://pypi.org/project/ag-os/) · [github.com/Manzela/Antigravity-OS](https://github.com/Manzela/Antigravity-OS)

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## Page 4 — Tasko AI + Seller App + Open-Source Index + Upstream OSS

### Tasko AI — production agentic system before the term existed | AI Product Engineer | Oct 2020 – Dec 2023

A WhatsApp-native AI errand assistant for paying clients, built on a four-layer agent stack (Classify → Retrieve → Execute → Verify) before the industry had a term for it. Trained from 21M+ real WhatsApp messages. 153 unique paying clients. 145 procedures, 67 formal protocols, 97 unique workflows, 30 metadata fields per task. Surface: [manzela.github.io/Manzela/tasko-ai/](https://manzela.github.io/Manzela/tasko-ai/).

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### Open-Source Projects

- **Antigravity-OS** — see Page 3. PyPI: `ag-os`. MIT-licensed. v1.4.0 (2026-05-08).
- **agent-dag-pipeline** — 7-node multi-agent DAG reference with Google ADK + Vertex AI + O-R-A-V evaluation + DPO flywheel. Apache-2.0. → [github.com/Manzela/agent-dag-pipeline](https://github.com/Manzela/agent-dag-pipeline)
- **gemma4-vllm-deployment** — forensic runbook of **20 distinct failure modes** deploying Gemma 4 26B-A4B-it MoE (128 experts, 4 active per token; 4B params per forward pass on a 26B model) on Vertex AI / vLLM. Includes the “unsolvable dependency triangle.” → [github.com/Manzela/gemma4-vllm-deployment](https://github.com/Manzela/gemma4-vllm-deployment)
- **WP-Multisite** — TNG WordPress multisite AI-search-aware infrastructure blueprint. → [github.com/Manzela/WP-Multisite](https://github.com/Manzela/WP-Multisite)
- **Atelier** — see Page 2. → [github.com/Manzela/Atelier](https://github.com/Manzela/Atelier)
- **AutonomousAgent** — production wrapper for Nous Research’s Hermes Agent with tiered sandboxing, OpenTelemetry, sops-encrypted secrets, network egress allowlist, Telegram approval gates. Phased path to Atropos RL training. → [github.com/Manzela/AutonomousAgent](https://github.com/Manzela/AutonomousAgent)

Active open-source contributor across `anthropics/claude-code`, `NousResearch/hermes-agent`, and `Agent-Field/agentfield`.

Daniel Manzela · v5.2 · generic · as of 2026-05-19